

Homework 5, due Sep 24 before/in class

1. At time $t = 0$ electron has velocity v_0 directed along y axis. There is electric field E_0 along z axis. Find $y(t)$ and $z(t)$ (fully relativistic).
2. Earth's magnetic field on the surface is approximately 1 Gauss. Earth magnetosphere extends approximately to 10 Earth radii, where it interacts with the solar wind, Fig. 1. Solar wind has a velocity $v_w \approx 500$ km/s. Find Larmor radii of protons from the Solar wind as they penetrate in the outer parts of the Earth magnetosphere.
3. Some laboratory device produces horizontal magnetic field of 10^3 G. A proton happens to be inside this device, Fig. 2. Find drift velocity of a proton induced by the gravity of Earth.

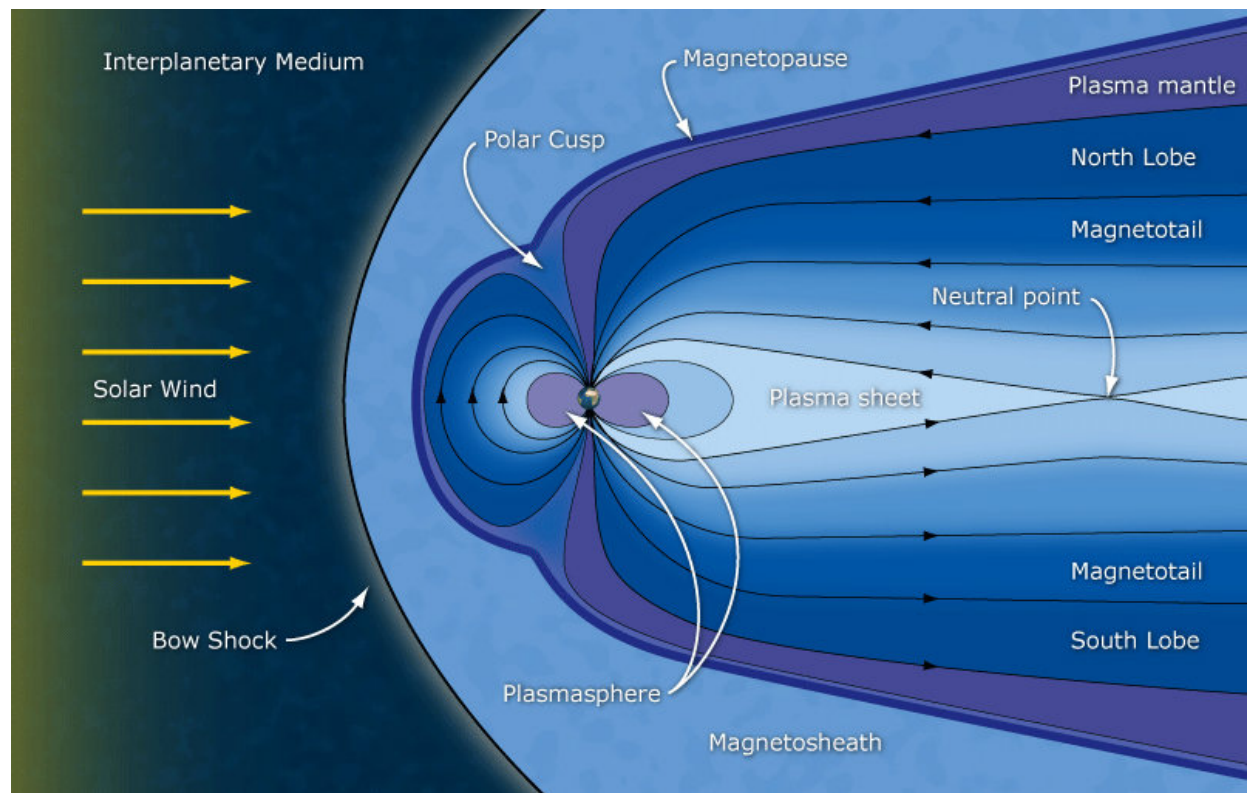


Fig. 1.— Magnetosphere of the Earth.

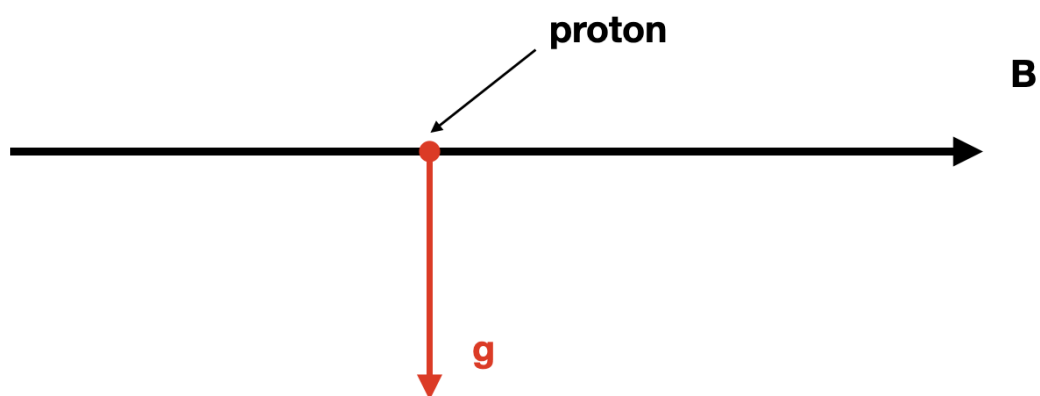


Fig. 2.— Magnetosphere of the Earth.